

Does your AI 3D model actually work?

What deterministic mesh checks can — and cannot — tell
you about generated GLB / OBJ / STL files

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modelfy.art · based on a 26-page pre-registered study



The short answer

Geometric validity is not perceptual quality.

Cheap CPU checks reliably catch injected holes, floaters and intersecting parts — but they barely track whether people think a mesh looks right.

Use checks as

validity gates

Is it printable, importable, riggable? Holes & floaters: AUROC 1.000 on injected defects.

Don't use checks as

quality judges

Correlation with human geometry ratings: $|r| \leq 0.2$ within generators.

Combine with

eyes or a VLM

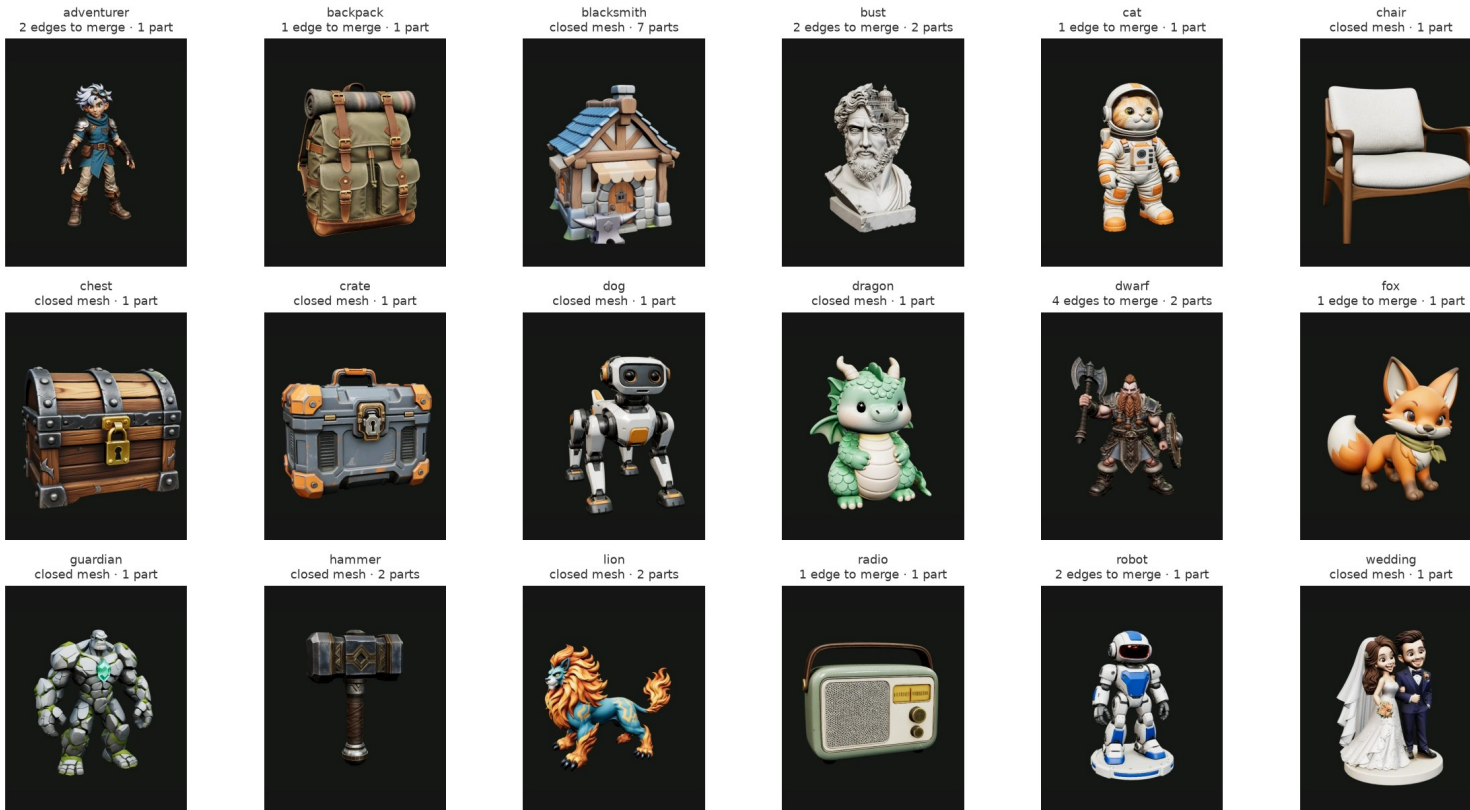
VLM judges scored MCC 0.335–0.341 vs 0.272 for the probe model; no significant paired difference.



WHY IT MATTERS

A textured render doesn't show the geometry

18 polished AI models (modelfy.art showcase) — what a 2-second geometry check reads



10 of 18 closed (watertight) as downloaded; 0 open holes in any model; the other 8 have 1–4 non-manifold edges to merge. Renders: modelfy.art homepage showcase. Checks: github.com/Stark-Will/geometric-probes-3d.

- ▶ AI image-to-3D output goes straight to 3D printers, game engines and AR viewers.
- ▶ Each destination cares about different things: closed edges, loose parts, wall thickness, inner layers.
- ▶ Even polished models are worth a quick check: **10 of 18** are closed (watertight) as downloaded; the other 8 need 1–4 non-manifold edges merged.
- ▶ A 2-second geometry check shows what the texture can't.

Six families of deterministic probes (30 features)

01

Open boundaries

edges used by only one face;
boundary loops & length

02

Components

loose parts / floaters; small parts <
1% of area

03

Self-intersection

faces that cut through other faces

04

Hidden surface

surface never visible from 64
outside directions

05

Thin walls

local thickness < 0.5% / 1% of
bbox diagonal

06

Roughness

dihedral angles, angle defect,
triangle quality

CPU only (Trimesh, PyMeshLab, Embree) · deterministic · ~2–3 s for a 150K-face mesh



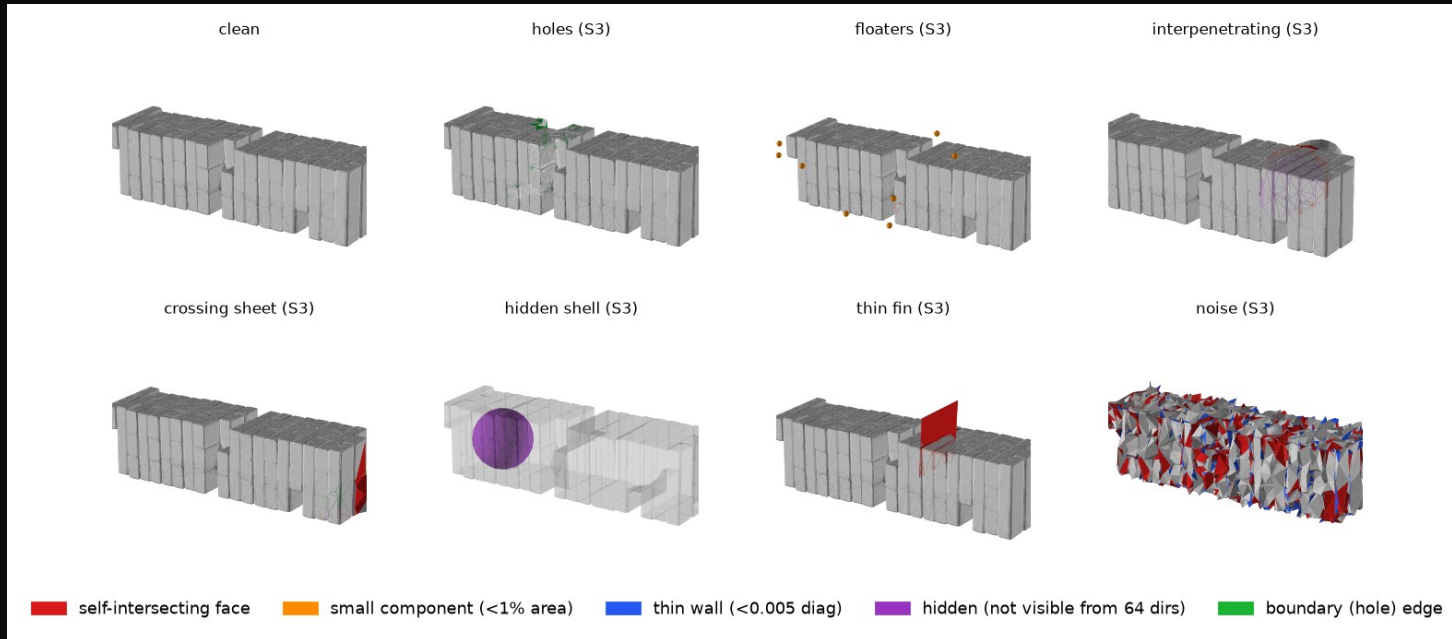
Four kinds of evidence, all pre-registered

Evidence	Dataset	Size	Question
Injected defects	Google Scanned Objects	120 real scans × 7 defect types × 3 severities	Can a check detect a known defect?
Expert labels	3D-DefectBench	678 generated GLBs; expert + crowd labels; 12 VLM judges' votes	Probes vs vision-language judges
Human ratings	MATE-3D	1,280 text-to-3D meshes with geometry MOS	Do checks track perceived geometry?
Scale-up	Hi3DBench	2,797 labelled meshes, MLLM pseudo-labels	Does the pattern hold at scale?

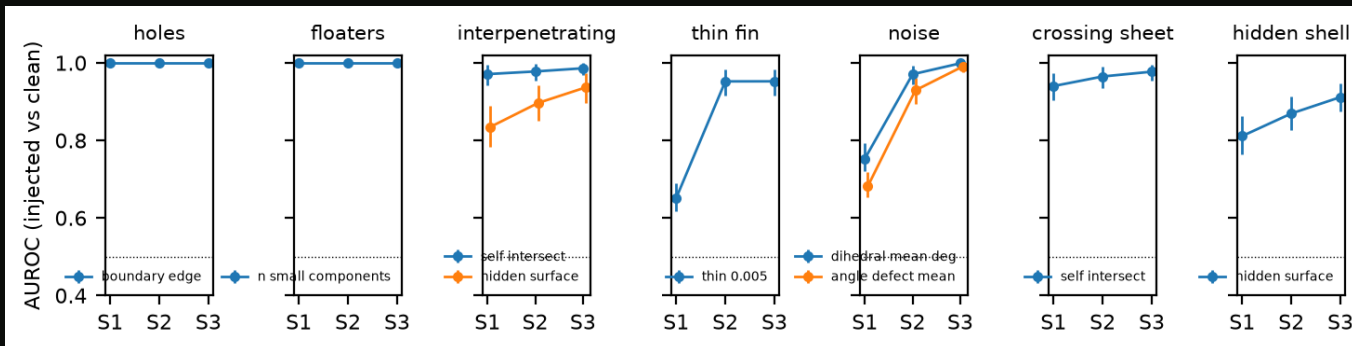
Analysis plans were committed before scoring; null and negative results are reported.

RESULT 1 · INJECTED DEFECTS

Known defects are caught — mostly

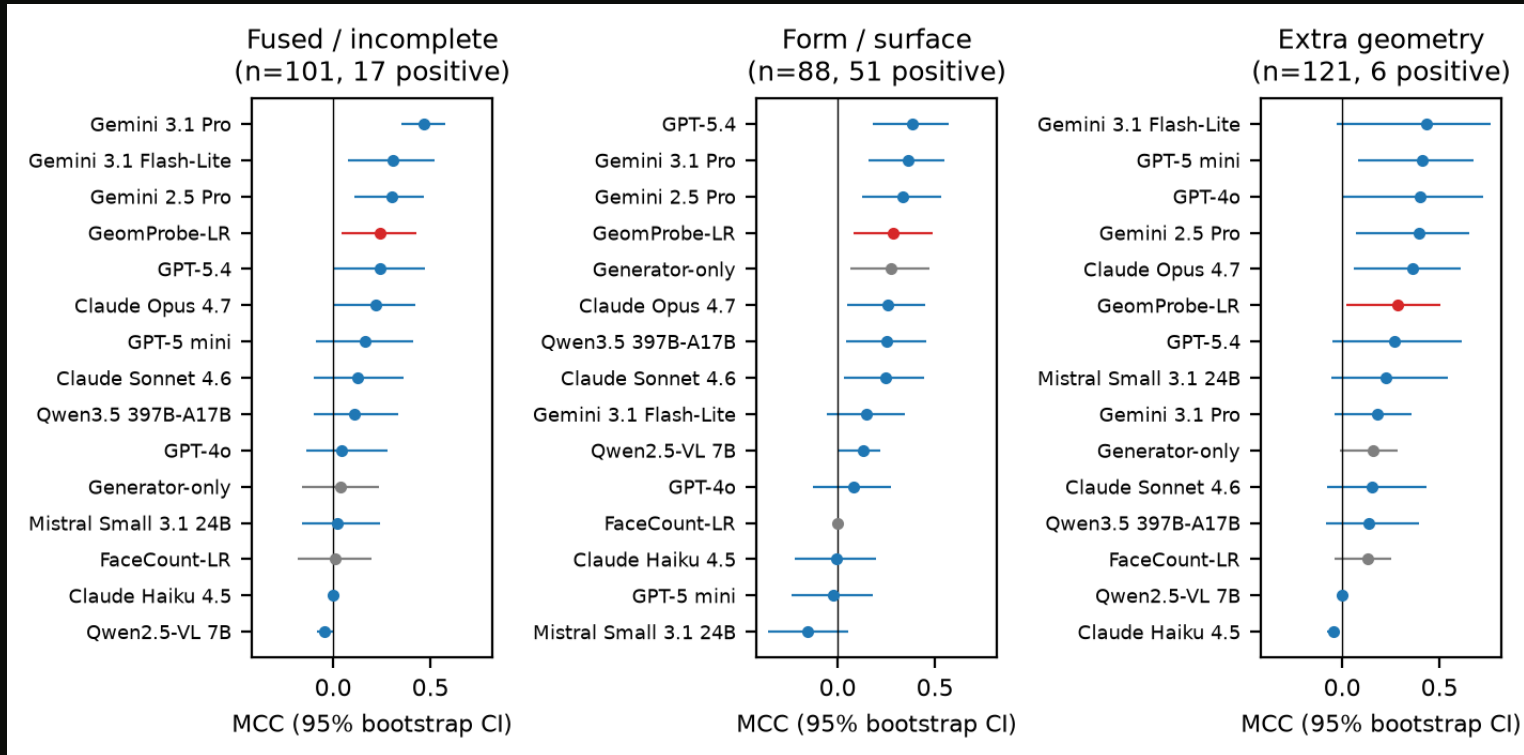


- ▶ Holes, floaters: **AUROC 1.000**
- ▶ Interpenetrating parts / crossing sheets: **0.940–0.986**
- ▶ Thin fins: **0.952** (two larger severities)
- ▶ Hidden shells: **0.811–0.912**
- ▶ Vertex noise: **0.753–0.999**
- ▶ **Note:** hidden-surface and self-intersection checks also respond to other defects (cross-talk).



RESULT 2 · EXPERT LABELS

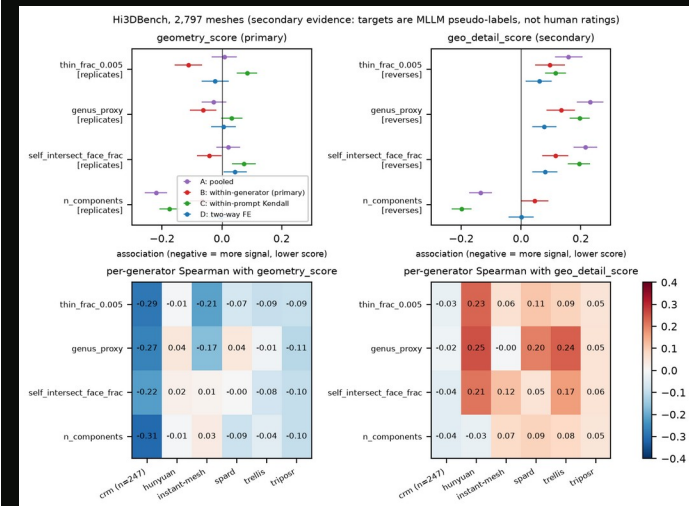
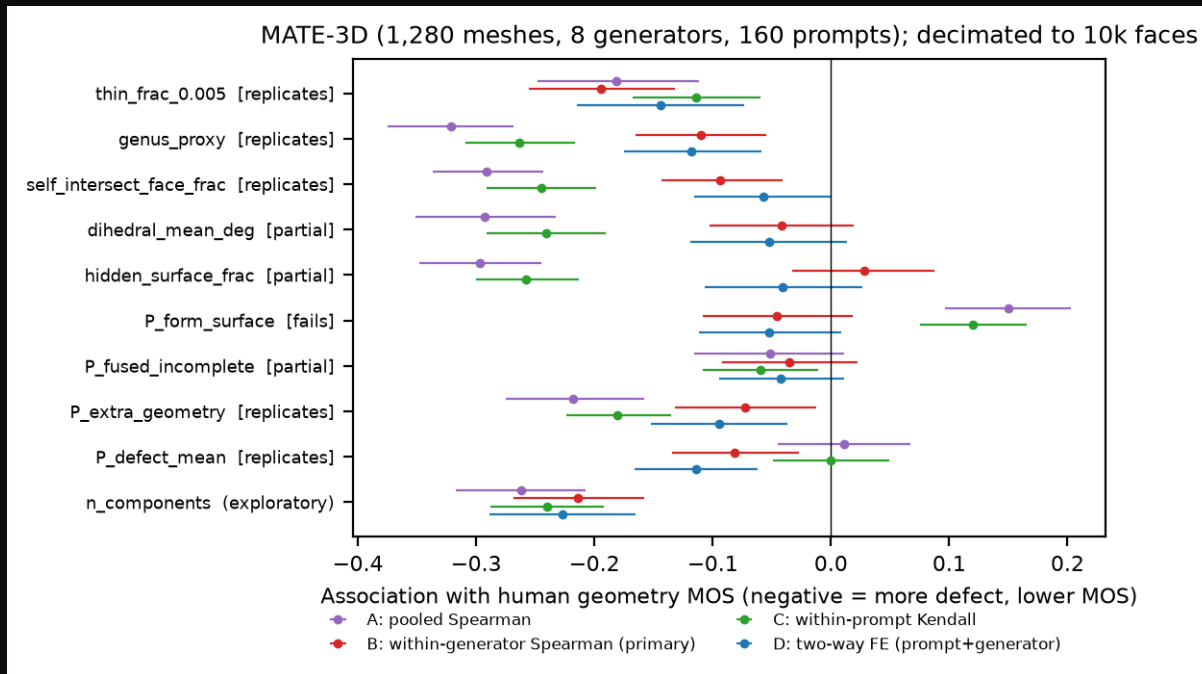
Against vision-language judges: close, not better



- Probe model (30 features, crowd-trained): **MCC 0.272**
- Best VLM judges: **MCC 0.335–0.341**
- No paired McNemar test vs the four strongest judges was significant.
- Form/surface performance was largely explained by generator identity — a confound, not skill.



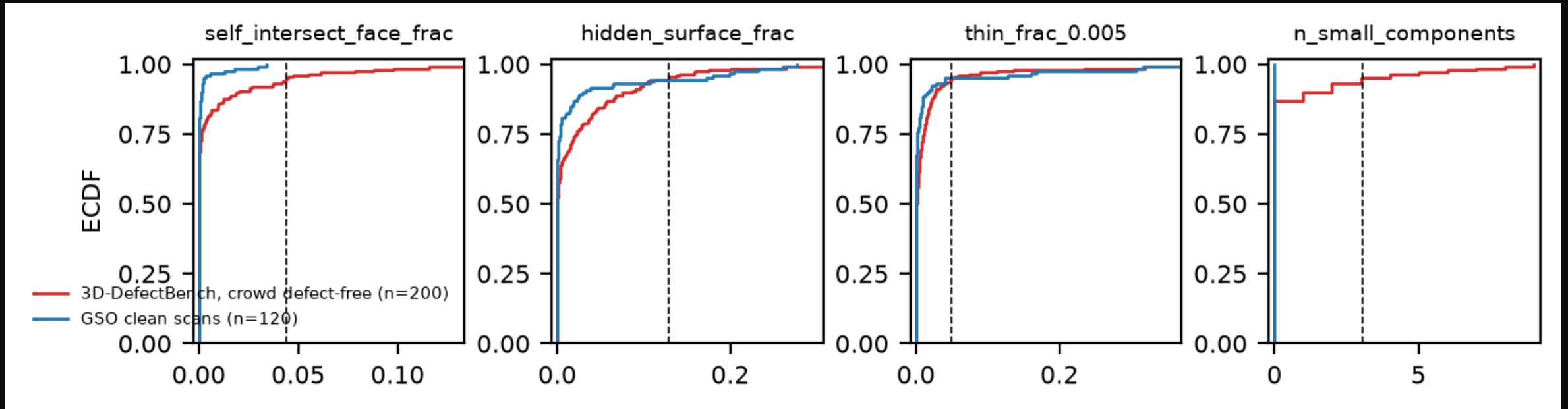
Checks barely track perceived geometry



- **MATE-3D: $|r| \leq 0.2$**
within generators
- **Hi3DBench (2,797): $|r| \leq 0.111$**
- Did not survive adjustment for prompt.



"Defect-free" assets are geometrically dirty — but watertight

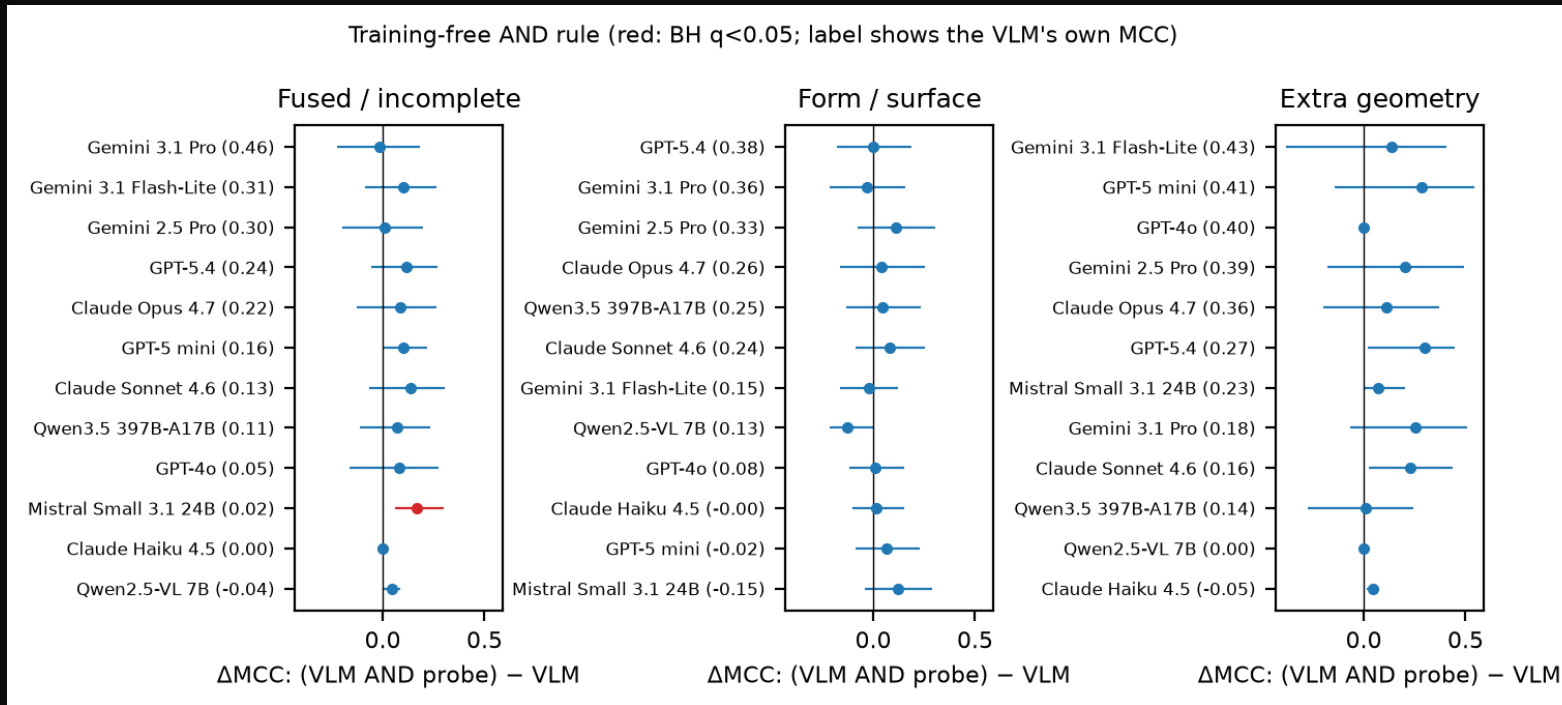


- **98.4% of the 678 generated GLBs are watertight.** Yet on 200 crowd-labelled defect-free assets the 95th percentile is 4.4% self-intersecting faces, 12.8% hidden surface and 5.0% thin-wall faces — far above clean scans.
- Strict zero thresholds also fire on undamaged real scans (**specificity 0.79 / 0.58 / 0.27**) → calibrate thresholds per destination; a non-zero reading is not automatically a failure.



RESULT 5 · COMBINING

Probes + VLM votes: small, fragile gains



- Training-free rule: flag only when a VLM vote AND a probe flag agree.
- MCC went up in **29 of 36** judge × defect pairs...
- ...but was significant in **only 1**.



TAKEAWAY

What this means in practice

Gate

Run checks before a mesh reaches a printer, engine or AR viewer. Holes and floaters are near-certain catches.

Calibrate

Set thresholds per destination (print $\neq$ game $\neq$ web). A single non-manifold edge may be harmless for a web viewer.

Look

Checks can't tell you if a chair looks like a chair. Keep a human (or a VLM) for appearance.

Control

Report generator identity and face count — they explain a lot of apparent 'quality'.



PART 2

Check your AI 3D model in 10 minutes

Worked examples: the 18 public showcase models on modelfy.art, read with the same checks

18 polished models, one command: how to read the output

10 / 18

closed (watertight) as
downloaded

0

open boundary loops in
any model

1–4

non-manifold edges to
merge in the other 8

2.0–2.6 s

CPU time per 150K-face
mesh

- ▶ Where a model is not closed, it comes down to 1–4 local non-manifold edges, not holes — quick to find in Blender (Select › All by Trait › Non Manifold).
- ▶ Detail shows up in the numbers: the blacksmith house reads 30.63% thin-wall faces (thicken before printing), the wedding topper 15.36% inner surface (simplify for print or games), the hammer 27 degenerate faces (merge).
- ▶ **Note:** curated showcase models, web-optimised versions — an educational example, not a benchmark.



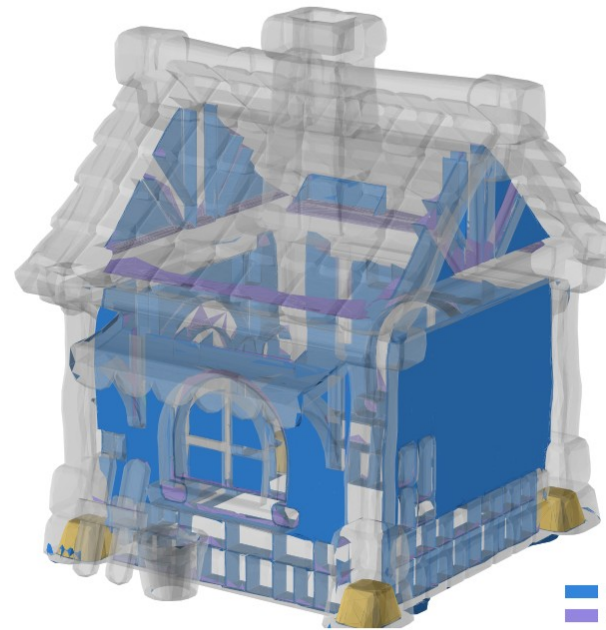
Thin walls to thicken before printing

What the checks show on a detailed model: thin walls to thicken before printing

modelify.art showcase render



geometry view · areas worth a look



Check readout — blacksmith
149,984 faces

closed (watertight): yes
open boundary loops: 0
non-manifold edges: 0
separate parts: 7 (small: 5)
self-intersecting faces: 0.005%
inner (hidden) surface: 11.67%
thin-wall faces: 30.63%
check time: 2.6 s (CPU)

Before printing:

- Thicken roof & awning plates (Solidify; check at print size)
- Join or remove 5 small parts

- thin wall (< 0.5% of bbox diagonal) — thicken for printing
- inner (hidden) surface — simplify for print / games
- small separate part (< 1% area) — join or remove
- self-intersection — merge or remesh
- open edge — fill

Mesh: modelify.art public homepage showcase (web-optimised GLB). Numbers: full-resolution mesh; overlay drawn on a 40k-face display copy.
Checks: github.com/Stark-Will/geometric-probes-3d (CPU-only, deterministic). Educational example, not a benchmark.



Inner layers to simplify for print or games

What the checks show on a layered model: inner surfaces to simplify for print or games

modelify.art showcase render



geometry view · areas worth a look



Check readout — wedding
150,462 faces

closed (watertight): yes
open boundary loops: 0
non-manifold edges: 0
separate parts: 1 (small: 0)
self-intersecting faces: 0.356%
inner (hidden) surface: 15.36%
thin-wall faces: 0.98%
check time: 2.5 s (CPU)

Before printing:

- On screen: nothing to change
- Print / games: simplify veil (e.g. voxel remesh)

- thin wall (< 0.5% of bbox diagonal) — thicken for printing
- inner (hidden) surface — simplify for print / games
- small separate part (< 1% area) — join or remove
- self-intersection — merge or remesh
- open edge — fill

Mesh: modelify.art public homepage showcase (web-optimised GLB). Numbers: full-resolution mesh; overlay drawn on a 40k-face display copy.
Checks: github.com/Stark-Will/geometric-probes-3d (CPU-only, deterministic). Educational example, not a benchmark.



What to check depends on where the model goes

3D printing (STL/3MF)

- ▶ Watertight: every edge on exactly 2 faces
- ▶ No loose parts unless intended
- ▶ Wall thickness at real scale $\geq$ your printer's minimum
- ▶ No self-intersections; consistent normals

Game engine (GLB/FBX)

- ▶ Face count fits the budget
- ▶ Few hidden inner shells (overdraw)
- ▶ Single-mesh parts that must move are separable
- ▶ Degenerate faces removed

Web & AR (GLB/USDZ)

- ▶ File size / compression (meshopt, WebP)
- ▶ Loads in a viewer without errors
- ▶ Scale & orientation correct (Y-up, metres)
- ▶ Texture maps present (baseColor/normal/ORM)

TROUBLESHOOTING

Symptom → cause → fix

Symptom	Likely cause	Fix
Not watertight	non-manifold or open edges	Blender: select Non Manifold › Merge by Distance › fill holes
Small separate parts	small disconnected components	delete parts < 1% area (trimesh mesh.split) or join intentionally
Walls too thin	single-sheet plates, fins	Solidify modifier / thicken, or scale up before printing
Inner (hidden) layers	layered cloth, overlapping shells	remove interior faces; voxel remesh for printing
Self-intersections	limbs/props fused through body	3D Print Toolbox › Intersections; boolean union or remesh
Won't open in a tool	meshopt / quantized web GLB	inspect with glTF-Transform; decompress before legacy tools

RESOURCES

Free tools & resources

Paper + code + per-asset stats

github.com/Stark-Will/geometric-probes-3d

Online 3D viewer (in-browser, GLB/STL/OBJ/USDZ...)

modelfy.art/3d-tools/online-viewer

3D format converter (GLB → STL / USDZ, OBJ → GLB...)

modelfy.art/3d-tools/file-converter

Image / text to 3D generator

modelfy.art

Blender 3D Print Toolbox · trimesh · PyMeshLab · glTF-Transform

open-source, free



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Disclosure: the author works on modelfy.art; no modelfy system was evaluated in the study.